

Chapter 2

Mid-IR Spectroscopy as a Tool for Cleanliness Validation

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1 BACKGROUND

Testing a surface for the presence of contaminants such as residual oils from a manufacturing process, residues of pharmaceutical or food products, or deliberately created hazards such as explosives, can be a considerable technical challenge. Available methods fall into two broad categories: direct methods such as visual inspection of the surface and indirect methods involving the quantitative removal and analysis of contaminants. In principle, direct methods are to be preferred if it can be shown that all of the target contaminants are detectable at the required surface concentration on the appropriate substrate.

The method discussed in this chapter is often referred to as infrared (IR) reflection-absorption spectroscopy (IRRAS). The principle is very simple: an IR signal is reflected from the target surface, either directly or at the so-called grazing angle. Depending on the nature of the substrate, the signal is either completely or partially reflected and travels through any layer of organic

contamination that is present on the surface, resulting in an absorption spectrum of the organic material. Assuming that the organic material has a mid-infrared (IR) spectrum (and almost all organic substances do), the contaminant spectrum can usually be identified and quantified on the basis of the IR spectrum.

For example, the presence of residual oils in the manufacturing of items such as sheet steel or airplane parts will sometimes interfere with later processes, such as painting. It was the need to evaluate aircraft wings prior to painting that led to the initial development of *in situ* grazing-angle mid-IR spectroscopy as a surface analysis tool. Other industries, such as semiconductor manufacture, involve the routine handling of highly reactive gases such as silanes and require high levels of verified cleanliness for gas-handling equipment.

Other applications have been focused on detecting potentially hazardous materials in sensitive environments, a major example being pharmaceutical or food processing equipment. Cleanliness verification and validation is a topic of particular interest in the pharmaceutical industry due to the possible dangers of carryover or cross-contamination between batches of different drugs, for example. Two recent articles give a useful overview of the history [1] and current recommended practice [2] for arriving at safe and practical limits for residues on critical surfaces [3]. The field of “homeland security” also covers several areas where real-time detection of hazardous materials is important. The application of IRRAS for the detection of explosives [4] and chemical warfare agents [5] is under evaluation.

A survey carried out on behalf of the Parenteral Drug Association (PDA) in 2008 [6] shows that an overwhelming percentage (96%) of responding companies uses a swabbing technique to measure active pharmaceutical ingredient (API) residues. For residues of cleaning agents, rinse sampling is somewhat more common than swabbing. Both of these methods depend on the ability to remove the contaminating material, either mechanically or by solution, for subsequent analysis. In other words, these methods must be classified as indirect. Calculations of the surface concentration based on the amount detected by analyzing the swab usually include a factor to account for possible incomplete recovery from the surface. A direct method that has been discussed in recent years is the visual estimation of residues [7]. Conceptually, the application of a spectroscopic method directly to the sample surface can be viewed as an extension of the visual method, and thus as a direct method. It has been used successfully to obtain spectra of monolayers on metals and on liquid surfaces [8 and references therein]. Grazing-angle reflectance accessories have been available for use with Fourier transform infrared (FTIR) spectrometers for a number of years, and the method is well understood. However, the sample size that can be accommodated in conventional grazing-angle accessories places a limit on the method’s usability for cleanliness validation, unless equipment can be disassembled, or removable coupons are evaluated [6]. In 2003, a report from the US Naval Facilities Engineering Service Center (NFESC) [9] confirmed that the method is suitable for evaluating the cleanliness of critical surfaces

to levels below $1.0 \mu\text{g}/\text{cm}^2$, and a design was developed for a cart-mounted system to be used in a plant [9, p. 18].

In another application, the detection of low levels of hazardous materials such as explosives and their precursors has become a major area of research for so-called homeland security applications around the world. The ability to detect low levels of explosives in real time on a range of surfaces, including common materials used for passenger luggage and commercial shipping containers, has assumed increasing importance in recent years [10–14].

The mid-IR reflectance technique described in this chapter was originally developed in collaboration with the aerospace industry [grazing-angle method, 15] and the sheet metal manufacturing industry [direct reflectance method, 16] for the detection and quantification of oils and other contaminants on large metal surfaces. It has subsequently been adapted for use in pharmaceutical cleanliness validation [17], and for the detection of trace amounts of explosives [10–14] and chemical warfare agents [18], as well as applications that involve working with glass and plastic surfaces in addition to metals [11].

2 PRINCIPLES OF GRAZING-ANGLE FTIR

A detailed discussion of the theory of reflection-absorbance spectroscopy and of grazing-angle FTIR is beyond the scope of this article. Good descriptions and discussions can be found elsewhere [8,19–21].

In brief, mid-IR reflectance spectroscopy at the so-called gazing angle is one of the most sensitive optical measurement techniques available. In a grazing-angle experiment, the IR beam approaches the sample surface at an incident angle of approximately 80° to normal. A good simple explanation of the effect is given by Hoffard [9] as follows:

Predominantly, improved sensitivity at grazing angles results from the polarization phenomenon of electromagnetic radiation (Hoffard References 6 and 7). The electric vector of all electromagnetic radiation contains two components—the p-component and s-component. For radiation contacting reflective surfaces, the s-component, perpendicular to the plane of reflection (parallel to the plane of the surface), undergoes a phase shift of approximately 180 degrees. The vector sum of the incident and reflected s-component is almost zero at the surface; thus, the two components cancel each other out. At grazing angles, the p-polarized component undergoes a phase shift at the surface from approximately 20 to 180 degrees, depending on the exact angle of incidence. At large incident angles, this phase shift is approximately 90 degrees (Hoffard Reference 7). The vector sum of the incident and reflected p-component now gives an intense electric field oriented perpendicular to the reflecting surface. When passed through a polarizing lens, the s-component of the reflected radiation can be filtered out and only the p-component is detected and converted to a spectrum. Additionally, at large angles of incidence, the infrared beam contacts the contaminant-laden surface

at an increased effective path length through the infrared-absorbing material. In accordance with Beer's Law of absorbance, this enhances the absorption, which results in a stronger FTIR "signal" of the contaminant (Hoffard Reference 8).

Grazing-angle spectroscopy on the laboratory scale is generally carried out using a polarized signal. As Hoffard points out, the enhancement of the signal at the grazing angle is largely attributable to the *p*-polarized component of the IR signal. The additional enhancement due to the increased sampling area on the surface is secondary, although it is often invoked as a simple conceptual explanation of the technique. However, when optical fibers are used to transmit the IR signal, polarization of the signal is lost. Polarization-retaining optical fibers are not available in the mid-IR range. The signal enhancement down to low levels of surface contamination still holds.

3 DESCRIPTION OF THE METHOD

A note about units: mid-IR reflectance methods directly measure the amount of contamination on the sample surface. Results are generally presented either in units of film thickness (e.g., micrometers, μm , or Angstroms, $\text{\AA}$) or in units of surface concentration (weight per unit surface area, e.g., $\mu\text{g}/\text{cm}^2$ or mg/ft^2). Results in this article are given in $\mu\text{g}/\text{cm}^2$, except when other authors are quoted, in which case the original units are used.

Grazing-angle FTIR analysis of a surface requires the FTIR signal to be introduced to the surface at or near the grazing angle of approximately 80° to normal. The reflected signal must then be captured onto a suitable detector for processing and quantification. The examination of large and/or inaccessible surfaces *in situ* requires the ability to bring a grazing-angle sampling device into contact, or very close proximity, with the sample surface. Two approaches to this have been used in recent years. Small portable or semiportable spectrometers have been fitted directly with grazing-angle accessories so that the entire apparatus may be held in proximity to the sample surface [22], and a hand-held grazing-angle analysis head has been developed for use with a fiber-optic link to the spectrometer [15] (see Fig. 2.1).

FTIR spectroscopy requires a blank or background spectrum, and this is best obtained using a dedicated clean background coupon. It is important to use a background coupon that is composed of the same material as the target surface, e.g., the background spectrum should be collected from stainless steel when a steel surface is to be analyzed, or from a glass coupon when a glass-lined reactor is under investigation. Thought should also be given to the surface finish of the background coupon, as compared to the target surfaces. Two papers [23,24] have investigated the effect of varying surface roughness on grazing-angle IRRAS spectroscopy. In the first, it is stated:

For samples prepared on surfaces of a given roughness, the results are independent of the choice of background finish [23].

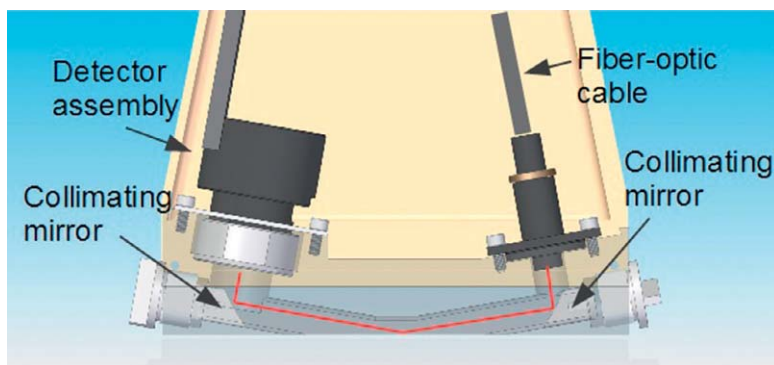


FIGURE 2.1 Schematic cross section of handheld grazing-angle head.

However, the overall conclusion is that increased surface roughness tends to degrade the predictive quality of chemometric calibrations and in particular to increase the errors associated with predictions. From a purely practical standpoint, a mirror-polished or uniformly smooth/matte background is the best choice. In the course of collecting data from different surfaces inside a reactor, for example, areas of variable roughness may be encountered and not all of these can be matched with a corresponding background coupon.

The calibration method used for the grazing-angle probe will depend on the nature of the analytical requirements. The method has been used successfully as a qualitative tool to identify low levels of contamination in pharmaceutical tanks based on simple visual evaluation of spectra (see details in the applications section below). A semiquantitative method based on comparative peak heights in the C-H stretch region of the spectrum ($2800\text{--}3000\text{ cm}^{-1}$) has been implemented in a different pharmaceutical setting, using a slim, direct reflectance mid-IR probe. Fully quantitative calibration of the method is carried out using chemometric methods, and several papers have been published on this topic [23–26].

The best method for preparing the calibration sample depends on both the identity and properties of the target compound(s), if known, and on the type of spectral interpretation or calibration that is required. If simple detection of organic material is all that is required, visual inspection of the grazing-angle spectrum will sometimes be sufficient. An example of this is shown in Fig. 2.2. Four grazing-angle IRRAS spectra were taken from the surface of a part removed from a pharmaceutical extruder after cleaning [27]. The surface was visually clean, but a clear spectroscopic signal was detected at one site (green (gray in print version) trace), a less clear one at a second site (navy (black in print version) trace), and very low or no signal was detected at two other sites (maroon (dark gray in print version) and pink (light gray in print version) traces). Visual comparison of the spectra, including the features near

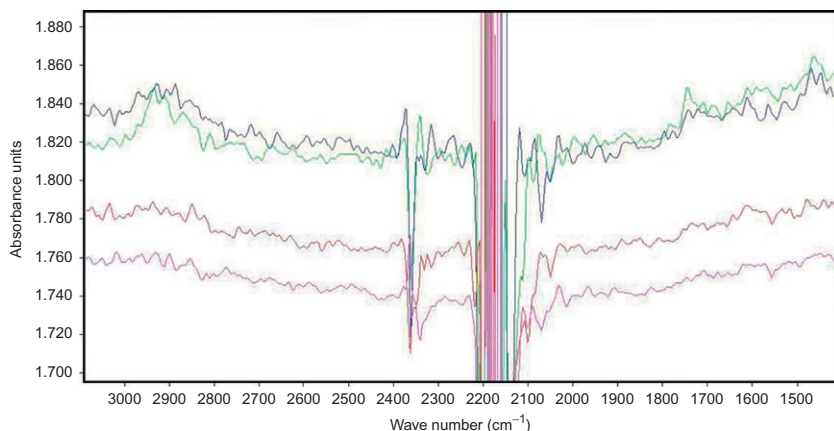


FIGURE 2.2 Grazing-angle IRRAS spectra from four sites on a piece of extrusion equipment.

2900 cm^{-1} and those in the fingerprint region between 2000 and 1400 cm^{-1} , strongly suggested that the contamination was caused by the residual cleaning material. A full quantitative study was later undertaken (Fig. 2.2).

Hamilton et al. [26] showed that IRRAS spectra can be successfully calibrated using a simple peak height measurement to below $1\text{ }\mu\text{g}/\text{cm}^2$ for exemplary pharmaceutical compounds, even on glass surfaces. However, at lower concentrations, they found that nonlinearities were common, and interference by peaks from atmospheric water limited the applicability of univariate calibrations at the lower surface concentrations that are often required for pharmaceutical cleanliness validation. Note that IRRAS is an open-path technique and the resulting spectra include features attributable to atmospheric water and carbon dioxide. Furthermore, real cleanliness validation exercises will often include two or more compounds on the target surface, so that the IRRAS intensity in any given region cannot be unequivocally assigned to just one component. Variations in surface finish can also have confounding effects on peak intensity. Multivariate modeling methods based on factor analysis and partial least squares (PLS) fitting have therefore been adopted for fully quantitative calibration of grazing-angle data. A series of papers from the University of Canterbury [24–26] explore the application of PLS methods to mixtures of compounds on glass and metal surfaces of varying roughness, demonstrating that this approach can give acceptable quantitative results well below $1\text{ }\mu\text{g}/\text{cm}^2$.

The problem of preparing adequate calibration standards is discussed at length by both Hamilton and Perston [20,21]. For demonstration purposes, and for qualitative analysis, simple smearing of a solution of the target compound onto a sample coupon, followed by drying, can provide an adequate sample. Fully quantitative samples of soluble materials can be prepared by aerosol spraying of solutions or mixtures onto coupons, followed by IRRAS sampling of the dry coupons, and finally by careful washing with a suitable solvent and

analysis of the washings by HPLC (high-pressure liquid chromatography), UV-vis (ultraviolet-vis) spectroscopy, or ^1H -NMR (for compounds not active in the UV). In the case of poorly soluble compounds which cannot be quantitatively washed from sprayed coupons, a method of quantitative dot-matrix printing is recommended [28].

4 ADVANTAGES AND DISADVANTAGES

IRRAS offers a number of advantages compared to other methods of surface analysis and cleanliness validation or verification.

4.1 Direct, Real-Time Method

The methods most commonly used for cleanliness verification and validation are swabbing and rinse analysis. Swabbing involves wiping a defined area of the surface with a solvent-wetted swab, then extracting the swab and analyzing the extract, usually using HPLC. The method is effective only for soluble materials, and a 2005 paper on swabbing method development [29] gave recovery levels that did not exceed 87%. When possible operator-to-operator variation is taken into consideration, swabbing can be characterized as a method with considerable weaknesses in the areas of recovery and reproducibility. There is also the potential for considerable delay in obtaining results, as a time-consuming laboratory analysis is required after the swabbing step is complete.

Rinse water analysis, often by UV-vis spectroscopy, has the potential to be carried out in near real-time and to be very sensitive. However, it measures only the material that is extracted from the surface and is limited to UV-active compounds (some classes of cleaning agent are not UV active).

Visual inspection of surfaces is a method that has attracted some interest recently [7]. Clearly, a vessel that fails a visual inspection will require additional cleaning. However, the method is hard to calibrate and is subject to considerable possible variation and error based on lighting conditions and operator training or experience. In published examples, a practical limit near $3\text{ }\mu\text{g}/\text{cm}^2$ is claimed. However, this could be expected to vary based on factors such as surface condition (roughness, etc.) and the nature of the contamination. A film of detergent, for example, would generally be more difficult to detect visually than a layer of a solid drug substance. Visual inspection does not offer any possibility of identifying the contaminant, should one be detected.

IRRAS shares the advantage of being a direct method and can be considered an extension of visual methods into the mid-IR region of the spectrum. Once calibrated, the method has the potential to return results within seconds of the measurement, and since it is faster and less laborious than swabbing, many more sites on a target surface can be sampled in a short time. Furthermore, even qualitative IRRAS results can be used to fully or partially identify any contaminants that are detected.

4.2 Detect Anything With An IR Spectrum

Mid-IR spectroscopy is very close to a universal detection method for organic compounds. Furthermore, it is usually possible to distinguish and quantify individual compounds in a mixture. For example, an API can be measured in the presence of other substances such as common excipients and cleaning materials [17,25].

4.3 Automation

An advantage of the chemometric approach to calibrating an IRRAS system is that the calibration models can be saved and implemented into the data collection package to give real-time data as the target surface is analyzed. This means that decisions can be made about, for example, reanalyzing problem areas, or extending investigation over additional surfaces, as the sampling is in progress. It is also possible to “map” a large area systematically and overlay or coordinate the results with other methods, such as visual inspection.

4.4 Access to Small Spaces/Flat Surfaces

A standard grazing-angle head, coupled to a spectrometer by a fiber-optic cable, samples an area of a few square centimeters at a time. This places limitations on the size and accessibility of parts and areas that can be sampled. The grazing-angle method can be complemented, or if necessary replaced, by a direct reflectance method using a slim (diameter 6 mm) probe that samples a much smaller area. This method is less sensitive, since it does not take advantage of the grazing-angle enhancement of the signal. Nonetheless, it is a useful technique for analyzing small or low-access surfaces, and an example of its use is given below.

5 RESULTS AND APPLICATIONS

5.1 Aerospace

The fiber-optic coupled IRRAS head was originally developed for use in the aerospace industry, to detect very low levels of contamination on large metal objects. Accordingly, the first grazing-angle heads developed for this application were quite large, although usable in handheld mode. For example, a fiber-optic couple head with a footprint of 38.1 cm by 6.4 cm, and an elliptical data collection area approximately 12 cm by 2 cm was used for early measurements. Lubricants such as Meropa (Chevron U.S.A. Inc., San Ramon, CA) were detected down to very low levels on metal surfaces, and simple visual inspection of spectra (Fig. 2.3) indicated that the method had the potential to be quantitative. Studies of silicone oil and hydrocarbon oil on aluminum

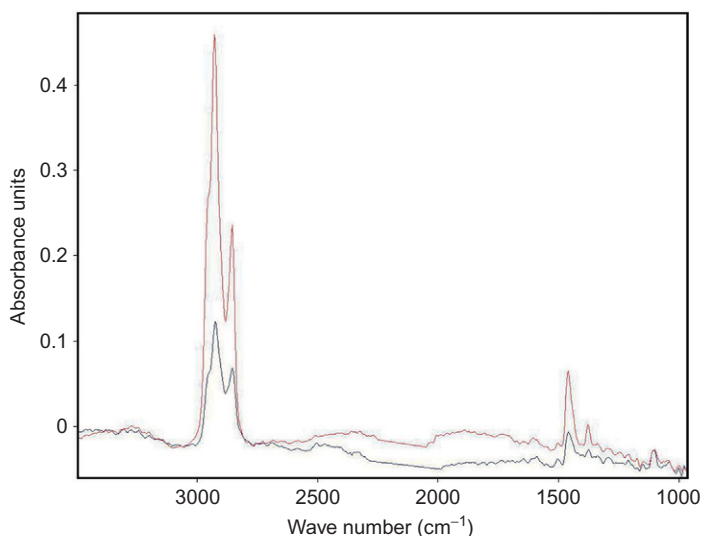


FIGURE 2.3 Handheld IRRAS spectra from an aluminum surface contaminated with 1 (navy (dark gray in print version)) and 4 (red (light gray in print version)) layers of 10% Meropa.

coupons [15] confirmed that these materials could be detected on aluminum surfaces at levels as low as $3.01 \mu\text{g}/\text{cm}^2$ ($2.8 \text{ mg}/\text{ft}^2$) using a grazing-angle IRRAS device.

In 2003, a report from the Naval Facilities Engineering Research Center (NFERC) [9] described a project to evaluate the use of grazing-angle mid-IR spectroscopy as a tool for online cleanliness verification at Department of Defense cleaning facilities, concentrating on univariate calibrations based on integrated peak areas in selected regions of the spectrum. Although the report mentions a detection limit as low as $0.22 \mu\text{g}/\text{cm}^2$, which is close to the levels established in the University of Canterbury work quoted above [24–26], the calibration curves that are shown, based on peak areas, are calculated in the range between 0.1 and $2.4 \mu\text{m}$ film thickness (equivalent to approximately $10\text{--}100 \mu\text{g}/\text{cm}^2$, assuming a density of $1 \text{ g}/\text{cm}^3$ for the contaminant). This is well above the surface concentration for which grazing-angle methods are usually employed. However, NFERC did develop a semiportable device for *in situ* evaluation of surface cleanliness on a variety of materials.

5.2 Manufacturing

Surface cleanliness is important in a wide range of industries. The equipment that is used to handle high-purity gases for semiconductor manufacture must be free of contamination for reasons of both quality control and safety. A grazing-angle IRRAS device that was specially built to examine the interior surfaces of ball-valve housings and other hollow objects has been in use to

check valves use in this industry for several years. The target contaminant is hydrocarbon oil and the method is a semiquantitative one, based on a threshold value for the height of a peak in the C-H stretch region of the spectrum. A cleanliness level of $<1 \text{ mg/m}^2$ ($<0.1 \text{ } \mu\text{m/cm}^2$) is being maintained [30].

5.3 Pharmaceutical Applications

An investigation carried out at the University of Puerto Rico, Mayaguez, in collaboration with Pfizer [31], demonstrated that detection of a pharmaceutical ingredient on aluminum surfaces was possible down to levels of $4 \text{ } \mu\text{g/cm}^2$ (the lowest practical level that was measurable using a simple gravimetric method). Crucially for future application in the pharmaceutical industry, a direct comparison was made between the IRRAS results and swabbing results obtained from the same coupons, using a standard industrial swabbing/HPLC method. The comparison of both IRRAS and swabbing results against the gravimetrically determined surface loading showed clearly that IRRAS is capable of at least the same level of residue detection and quantification as the widely used swabbing method (Fig. 2.4). The same paper also established that the average errors associated with a univariate IRRAS calibration based on peak heights were comparable to the errors in the swabbing results.

A series of papers and two dissertations from the University of Canterbury [20,21,23–26] explored the practicality of using the grazing-angle method on

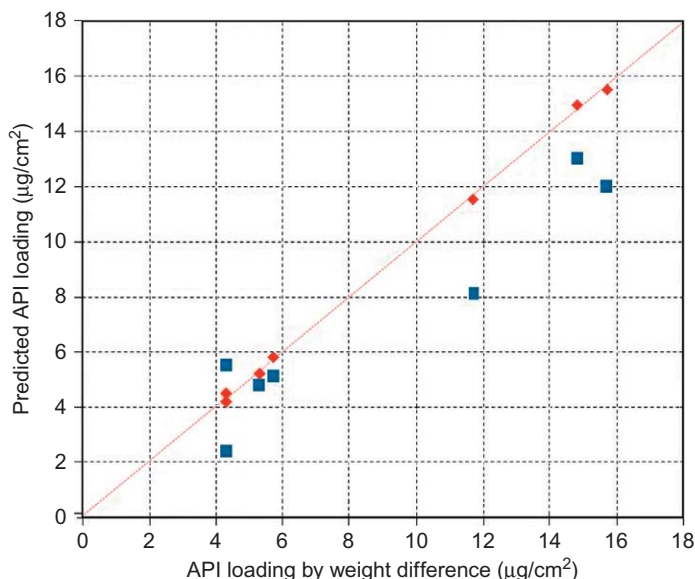


FIGURE 2.4 Comparison of IRRAS (red diamonds (light gray in print version)) and swabbing (blue squares (dark gray in print version)) results for surface analysis.

metal and glass surfaces, and discussed the effect of variations in surface roughness and of the presence of more than one compound on the surface. As a result of this work, it is apparent that

- a single compound can be measured on glass or metal with a prediction error (root mean square error of cross-validation in a partial least squares regression method) in the region of $0.1 \mu\text{g}/\text{cm}^2$.
- the presence of variable amounts of water vapor in the environment can be accounted for by the chemometric method, and this is a better approach than precorrecting the spectra for water vapor.
- work done using an exemplary cleaning agent (sodium dodecyl sulfate, SDS) indicates that it can be detected and may exhibit lower prediction errors compared to other classes of compounds, possibly due to superior surface wetting, resulting in more uniform coating of calibration samples.
- the work with SDS incidentally demonstrates the utility of an IR spectroscopic method for detecting compounds that are not UV active.
- rougher substrate surfaces result in increased prediction errors. However, it is possible to develop acceptable chemometric calibrations that include samples of varying roughness and which can be used to examine a range of sample surfaces.

More recently, Nakagawa et al. [32] have reexamined the question of chemometric calibration methods for IRRAS as a means of pharmaceutical cleanliness validation. Building on the work at the University of Canterbury, they described a method called locally weighted partial least squares (LW-PLS) which was developed and tested using an exemplary drug substance (ibuprofen) and a typical excipient (magnesium stearate). The advantage claimed for LW-PLS versus conventional PLS and locally weighted regression (LWR) methods is the ability to accommodate nonlinearity and clustering in the calibration data. Briefly, in LW-PLS, the samples stored in the calibration set are weighted on the basis of distance from the query, and then used to build a calibration model. This enables the use of a model that can be optimized for each individual query and may facilitate life-cycle management of the calibration model.

5.4 Explosives and Chemical Warfare Agents

Beginning in 2004, Hernández-Rivera et al. at the University of Puerto Rico, Mayaguez, have published a series of papers on the use of fiber-optic grazing-angle IRRAS to study the presence and behavior of two classes of compounds: explosives [10–14] and simulated chemical warfare agents [5,18]. The method has allowed them to study not only the detection and quantification of these materials on a variety of surfaces, including metals and plastics, but it has also enabled investigation of degradation products and of the residence time of volatile materials (e.g., the explosive triacetone triperoxide) on different surfaces.

5.5 Tank Trials

In a recent trial at a pharmaceutical company [27], a grazing-angle IRRAS system was used to examine the interior surfaces of two recently cleaned production tanks and selected interior surfaces of the recovery tank and pump used during clean-in-place (CIP) procedures on the two tanks. An operator was lowered into the tank carrying the FTIR spectrometer, and then the IRRAS head and fiber-optic cable were lowered separately and connected *in situ*. The computer was operated by a second person outside the tank. A total of 69 spectra were collected over a period of 3 days from selected sites on the walls of the tanks, agitator shafts, and spray balls, and from the manways and manway covers. Separately, spectra were collected from surfaces of the CIP recovery tank and pump.

For example, Fig. 2.5 shows grazing-angle spectra collected from three areas inside the tank [27]. Two of the spectra are essentially featureless (the noise feature between about 2100 and 2200 cm^{-1} is due to impurities in the mid-IR optical fiber cable that was used). The third spectrum shows two characteristics typical of contaminated surface:

1. The baseline is shifted, in this case upward, due to a change in the surface reflectivity compared to the clean background coupon.
2. A distinct multippeak feature between 2800 and 3000 cm^{-1} and a broad feature centered at 3300 cm^{-1} indicate the presence of C-H functionality and liquid water, respectively.

Visual inspection of the spectra as they were collected was sufficient to identify the presence of organic contamination at one of the three sample locations.

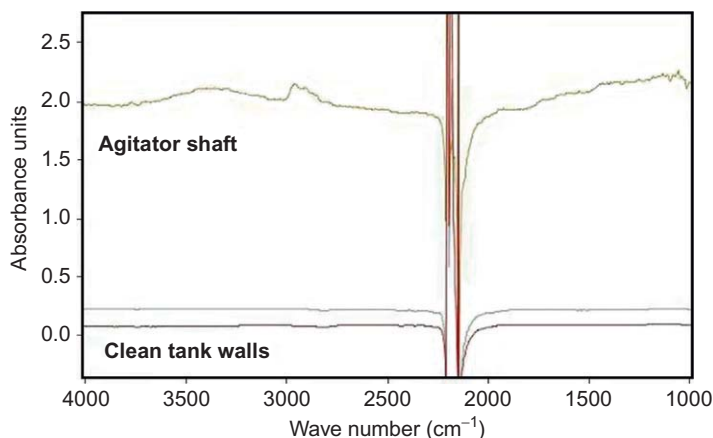


FIGURE 2.5 Comparison of IRRAS spectra from clean and contaminated surfaces inside a stainless steel tank.

Figure 2.5 shows one IRRAS spectrum taken from the surface of an agitator shaft inside one of the tanks. For comparison, two spectra from clean wall regions of the same tank are shown.

5.6 IRRAS by Direct Reflectance

As the grazing-angle approach to IRRAS has developed, it has become clear that there are some significant limitations to the method. First, even though successive generations of the handheld grazing-angle head have been designed for increasing compactness, it is still too large to be used for sampling small objects or low-access areas inside pharmaceutical reactors. It is sometimes possible to design dedicated grazing-angle devices for specific applications, such as the one described above that is used for hollow valve fittings. However, the size restriction is still significant.

Another concern is that the calibration of grazing-angle spectra can become problematic if a wide range of concentrations is included. For example, a very good calibration might be constructed for concentrations between 0.1 and $3 \mu\text{g}/\text{cm}^2$, with anomalies arising at higher loadings. However, in practice, the limit of visible detection for many materials on metal surfaces has been shown [7] to be near or above $3 \mu\text{g}/\text{cm}^2$. For completeness, the IRRAS calibration needs to be robust across wider range. This problem is addressed by Perston [21] and more recently Nakagawa et al. [32] have proposed a possible approach, which was discussed briefly above.

Both of these concerns can be addressed to some extent by recognizing that the IRRAS method need not necessarily operate at the grazing angle. It is the need to accommodate the required incidence angle of close to 80° that prevents the development of a smaller analysis head. One possible approach is to use a fiber-optic cable with a slim direct reflectance probe that can be as little as 6 mm in diameter. While this configuration is less sensitive than the grazing-angle IRRAS (by approximately $10\times$ in trials), it can be very useful for selected applications.

Some early work on this type of reflectance probe was done at the Institute for Production Engineering at the Saarland University, Germany [16]. The objective was to detect and measure residual oil on the surface of sheet metal and on smaller, more complex structures such as armatures.

Figure 2.6 shows a simple graph of absorbance in the C-H stretch region versus film thickness. There is considerable uncertainty associated with this calibration, which is based on a small amount of data. However, subsequent work with larger data based on both spray-coated and printed samples and using chemometric techniques has confirmed that the method can be used quantitatively and may be a more practical option for contamination levels above about $3 \mu\text{g}/\text{cm}^2$, or for small, poorly accessible surfaces.

Recent work on the detection of polymer residues on extrusion equipment has confirmed that the direct IRRAS method can be used semiquantitatively for

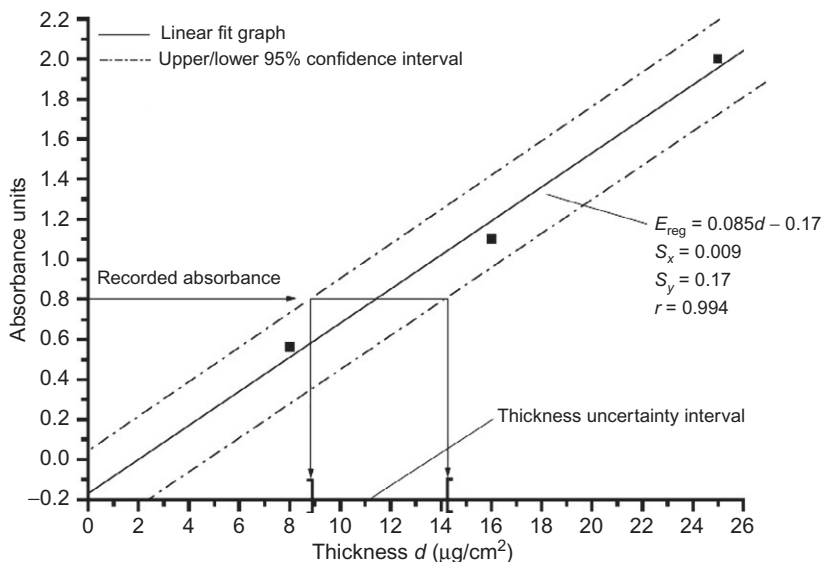


FIGURE 2.6 Graph of C-H absorbance against film thickness [16]. E_{reg} is the intensity of the peak used for the calibration; S_x and S_y are the intercepts of the best-fit line with the x - and y -axis, respectively; and r is the regression coefficient.

compounds with a strong, easily recognized signal, in this case a C-H stretch centered near 2900 cm^{-2} . Additional work with this particular material [28], which is only partially soluble, has led to the development of quantitatively printed coupons, prepared using an extract from the polymer. Chemometric calibrations based on data from these coupons show considerable promise for quantitative use of the direct reflectance IRRAS method at surface concentrations in the range from 3 to $20\text{ }\mu\text{g/cm}^2$.

6 FUTURE DEVELOPMENTS

As with many analytical methods, work with IRRAS, both at the grazing angle and using direct reflectance, continues to focus on the development of practical, robust calibrations. The fiber-optically coupled versions of this technology are already proving useful for real-time detection of low levels of organic contamination in a variety of different settings. As the technology is adopted more widely, possible areas for additional progress include

- integration with smaller FTIR units.
- design of more compact grazing-angle heads (bearing in mind the trade-off in terms of sensitivity).
- improvement of software interfaces to enable routine use in manufacturing settings.

7 SUMMARY

Mid-IR IRRAS, both at the grazing angle and in the direct reflectance mode, is a useful tool for identification and measurement of organic contamination on a range of surfaces, including metals and glasses. By using fiber-optics to connect a handheld reflectance head to an FTIR unit, it is possible to use the IRRAS method as an *in situ*, real-time tool for cleaning verification and validation. As with many analytical methods, calibration is an important part of successful implementation. A good deal of work has been done to demonstrate a full range of qualitative, semiquantitative, and fully quantitative applications of IRRAS in surface concentration ranges that are relevant for pharmaceutical cleaning validation in particular.

REFERENCES

- [1] A. Walsh, Cleaning validation for the 21st century: acceptance limits for active pharmaceutical ingredients (APIs): part I, *Pharm. Eng.* 31 (2011) 74.
- [2] A. Walsh, Cleaning validation for the 21st century: acceptance limits for active pharmaceutical ingredients (APIs): part II, *Pharm. Eng.* 31 (2011) 44.
- [3] S.L. Prabu, T.N.K. Suriya Prakash, R. Thirumurugan, Cleaning validation and its regulatory aspects in the pharmaceutical industry, in: R. Kohli, K.L. Mittal (Eds.), *Developments in Surface Contamination and Cleaning*, vol. 7, Elsevier, Oxford, UK, 2015.
- [4] O.M. Primera-Pedrozo, L. Pacheco-Londoño, O. Ruiz, Y.M. Soto-Feliciano, L.F. de la Torre-Quintana, S.P. Hernández-Rivera, Characterization of thermal inkjet technology TNT deposits by fiber optic-grazing angle probe FTIR spectroscopy, in: E.M. Carapezza (Ed.), *Proceedings of SPIE 5778, Sensors and Command, Control, Communications, and Intelligence (C3I) Technologies for Homeland Defense IV*, SPIE, Bellingham, WA, 2005, pp. 543–552.
- [5] O. Ruiz-Pesante, L.C. Pacheco-Londoño, C. Leonardo, O.M. Primera-Pedrozo, W. Ortiz, Y. M. Soto-Feliciano, D.E. Nieves, M.L. Ramirez, S.P. Hernández-Rivera, Detection of stimulants and degradation products of chemical warfare agents by vibrational spectroscopy, in: A.W. Fountain (Ed.), *Proceedings of SPIE 6554, Chemical and Biological Sensing VIII*, SPIE, Bellingham, WA, 2007, pp. 65540B/1-65540B/10.
- [6] D. Leblanc, PDA survey results: cleaning validation sampling practices, *PDA Lett.* 44 (2008) 11.
- [7] R.J. Forsyth, Do visible residue limits make the 10-ppm carryover limit obsolete? *Pharm. Technol.* 34 (2010) 60.
- [8] J. Umemura, Reflection-absorption spectroscopy of thin films on metallic substrates, in: J. M. Chalmers, P.R. Griffiths (Eds.), *Handbook of Vibrational Spectroscopy*, vol. 2, John Wiley & Sons, Chichester, UK, 2002, pp. 983–998.
- [9] T.A. Hoffard, Grazing-angle Fourier Transform Infrared Spectroscopy for Surface Cleanliness Verification, Technical Report TR-2217-SHR, Department of Defense, Naval Facilities Engineering Service Center, Port Hueneme, CA, 2003.
- [10] O.M. Primera-Pedrozo, Y.M. Soto-Feliciano, L.C. Pacheco-Londoño, S.P. Hernández-Rivera, Fiber optic-coupled grazing angle probe-Fourier transform reflection absorption infrared spectroscopy for analysis of energetic materials on surfaces, in: O.J. Rees (Ed.), *Fourier Transform Infrared Spectroscopy: Developments, Techniques and Applications*, Nova Science Publishers, Hauppauge, NY, 2011, pp. 79–99.

- [11] O.M. Primera-Pedrozo, N. Rodríguez, L. Pacheco-Londoño, S.P. Hernández-Rivera, Detection of 2,4,6-trinitrotoluene on non-traditional surfaces using fiber optic coupled grazing angle probe: FTIR, in: B.F. Andresen, G.F. Fulop, P.R. Norton (Eds.), *Proceedings of SPIE 6542, Infrared Technology and Applications XXXIII*, SPIE, Bellingham, WA, 2007, pp. 65423J/1-65423J/10.
- [12] L.C. Pacheco-Londoño, O.M. Primera-Pedrozo, S.P. Hernández-Rivera, Evaluation of samples and standards of energetic materials on surfaces by grazing angle-FTIR spectroscopy, in: O.J. Rees (Ed.), *Fourier Transform Infrared Spectroscopy: Developments, Techniques and Applications*, Nova Science Publishers, Hauppauge, NY, 2011, pp. 159–175.
- [13] L.C. Pacheco-Londoño, A. Santiago, J. Pujols, O.M. Primera-Pedrozo, A. Mattei, W. Ortiz, O. Ruiz, M.L. Ramirez, S.P. Hernández-Rivera, Characterization of layers of tetryl, TNB, and HMX on metal surfaces using fiber optics coupled grazing angle-FTIR, in: B. F. Andresen, G.F. Fulop, P.R. Norton (Eds.), *Proceedings of SPIE 6542, Infrared Technology and Applications XXXIII*, SPIE, Bellingham, WA, 2007, pp. 65423K/1-65423K/10.
- [14] S.P. Hernández-Rivera, L.C. Pacheco-Londoño, J.R. Castro-Suarez, O.M. Primera-Pedrozo, H. Felix-Rivera, Remote Raman and infrared spectroscopy detection of high explosives, in: T.J. Janssen (Ed.), *Explosive Materials: Classification, Composition and Properties*, Nova Science Publishers, Hauppauge, NY, 2011, pp. 231–258.
- [15] P.J. Melling, P.H. Shelley, spectroscopic accessory for examining films and coatings on solid surfaces, U.S. Patent 6,310,348, 2001.
- [16] S. Behrning, H. Bley, A flexible method for detecting and quantifying organic films on sheet metal, in: J.R. Dufluo, M. Geiger, H.J.J. Kals, B. Shirvani, U.P. Singh (Eds.), *Sheet Metal 2001 Proceedings of the Ninth International Conference on Sheet Metal*, Katholieke Universiteit Leuven, Belgium, 2001, pp. 451–458.
- [17] K.B. Reddy, N. Teelucksingh, Quantification of active pharmaceutical ingredients on metal surfaces using a mid-IR grazing-angle fiber optics probe—an in-situ cleaning verification process, *Spectroscopy* 20 (2005) 16.
- [18] S.P. Hernández-Rivera, L.C. Pacheco-Londoño, O.M. Primera-Pedrozo, O. Ruiz, Y.M. Soto-Feliciano, W. Ortiz, Vibrational spectroscopy of chemical agents simulants, degradation products of chemical agents and toxic industrial compounds, *Int. J. High Speed Electron. Syst.* 17 (2007) 827.
- [19] M. Claybourne, Mid-infrared external reflection spectroscopy, in: J.M. Chalmers, P. R. Griffiths (Eds.), *Handbook of Vibrational Spectroscopy*, vol. 2, John Wiley & Sons, Chichester, UK, 2002, pp. 969–981.
- [20] M. Hamilton, Application of Grazing-Angle Reflection Absorption Fourier Transform Infrared Spectroscopy to the Analysis of Surface Contamination, Ph.D. Thesis, University of Canterbury, Christchurch, New Zealand, 2007.
- [21] B.B. Perston, Infrared Reflection-Absorption Spectrometry and Chemometrics for Quantitative Analysis of Trace Pharmaceuticals on Surfaces, Ph.D. Thesis, University of Canterbury, Christchurch, New Zealand, 2006.
- [22] M.T. Beecroft, M.M. Szczesniak, B.J. Smith, J.F. Matsumoto, J.P. Ferguson, Hand-held Spectrometer, U.S. Patent 7,236,243, 2007.
- [23] M.L. Hamilton, B.B. Perston, P.W. Harland, B.E. Williamson, M.A. Thomson, P.J. Melling, Fiber-optic infrared reflection absorption spectroscopy for trace analysis on surfaces of varying roughness: sodium dodecyl sulfate on stainless steel, *Appl. Spectrosc.* 60 (2006) 516.
- [24] B.B. Perston, M.L. Hamilton, P.W. Harland, M.A. Thomson, P.J. Melling, B.E. Williamson, Fiber-optic infrared reflection absorption spectroscopy for trace analysis on surfaces of varying roughness. Part II: acetaminophen on stainless steel, *Appl. Spectrosc.* 62 (2008) 312.

- [25] B.B. Perston, M.L. Hamilton, B.E. Williamson, P.W. Harland, M.A. Thomson, P.J. Melling, Grazing-angle fiber-optic Fourier transform infrared reflection-absorption spectroscopy for the in situ detection and quantification of two active pharmaceutical ingredients on glass, *Anal. Chem.* 79 (2007) 1231.
- [26] M.L. Hamilton, B.B. Perston, P.W. Harland, B.E. Williamson, M.A. Thomson, P.J. Melling, Grazing-angle fiber-optic IRRAS for in situ cleaning validation, *Org. Process Res. Dev.* 9 (2005) 337.
- [27] M.A. Thomson, Unpublished Results, 2010.
- [28] M.A. Thomson, Calibration of in-situ IRRAS analysis of steel surfaces using printed sample coupons, in preparation.
- [29] P. Yang, K. Burson, D. Feder, F. Macdonald, Method development of swab sampling for cleaning validation, *Pharm. Technol.* 29 (2005) 84.
- [30] Process and Inspection, Tong Kwang Valve Industrial Co. Ltd., New Taipei City, Taiwan, ROC. Available from: http://www.tk-valve.com/index.php?option=com_content&view=article&id=136&Itemid=62
- [31] N.K. Mehta, J. Goenaga-Polo, S.P. Hernández-Rivera, D. Hernandez, M.A. Thomson, P. J. Melling, Development of an in-situ spectroscopic method for cleaning validation using a mid-IR grazing-angle fiber optic probe, *BioPharm Int.* 15 (2002) 36.
- [32] H. Nakagawa, T. Tajima, M. Kano, S. Kim, S. Hasebe, T. Suzuki, H. Nakagami, Evaluation of infrared-reflection absorption spectroscopy measurement and locally weighted partial least-squares for rapid analysis of residual drug substances in cleaning processes, *Anal. Chem.* 84 (2012) 3820.